Summary Report of the Candidate Technical Challenge: Screening Novel Consumer Molecules

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1. Purpose:

Use Python + DeepPurpose to screen a curated ingredient list and identify molecules with strong predicted binding to consumer targets. Demonstrate justified target selection, a working script, and a prioritized shortlist for product use.

2. Results:

a) Target selection:

The three protein targets selected were BDNF (P23560), TAS2R38 (P59533), and ADORA1 (P30542). Each is a compact human protein with fewer than 400 amino acids, which reduces modeling computation and simplifies cloning and cell-based assays, enabling faster and lower-cost validation. Additionally, these targets are directly linked to consumer-perceptible outcomes (mood, taste, and alertness), allowing rapid progression from in-silico prioritization to benchtop validation and small human pilots.

b) Molecule prioritization:

Molecules were retained only if they ranked in the top 25 for all three targets. To preserve consumer suitability, compounds with anabolic/prohormone, hormonal, or antihistamine profiles (e.g., hexadrone, epistane, DHEA, cyproheptadine) were excluded. For each retained compound, a consensus mean rank was calculated as the average of its per-target ranks (lower = better). The eight prioritized compounds are beta-zeacarotene, schizandrol-A, boswellic acid, stigmasterol, hydrastine, egg yolk lecithin, berberine HCl, and berberine sulfate. This selection reflects cross-target strength and practical feasibility: carotenoids, sterols, and lecithin that have established safety and formulation paths; berberines show consistent signal but require a bioavailability plan; and hydrastine warrants additional safety review.

c) Insights, assumptions, next steps:

The consensus mean rank was used to normalize differences in score scales and to reward cross-target consistency. This produced a robust, easily interpretable order that prioritized compounds performing well across all three targets rather than focusing on a single model. Crucially, the ranking is conditional on the chosen target panel (BDNF, ADORA1, TAS2R38). Hence, different, non-redundant panels would change the priority order, suggesting that target group selection should reflect the desired product outcome.

DeepPurpose scores are comparative (not calibrated Ki), so rank-based decisions are appropriate. BDNF is treated as an expression/secretion readout with serum/plasma ELISA endpoints. Regulatory limits and TAS2R38 genotype effects (PAV vs AVI) are handled downstream.

To strengthen and operationalize these results, in the next steps:

- Dataset improvements: library cleaning + collapsing salt forms to prevent featurization failures and make parent-level comparisons consistent.
- Model upgrades: benchmark multiple DeepPurpose encoders with a light ensemble and uncertainty to reduce model bias and highlight stable leads.
- Evaluation & selection: run sensitivity checks to the target panel and the rank rule to ensure the ordering isn't an artifact of panel choice or a single scoring scheme.
- Execute small, genotype-stratified human pilots to estimate effect sizes and variability in the intended population.
- Plan practical formulations (e.g., microencapsulated carotenoids, oral-contact formats) to ensure deliverability and exposure.